

Same Meaning, Different Answer?

Evaluating Answer Consistency in Large Language Models Across Linguistic Transformations

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Introduction & Research Gap

Background. LLMs are usually tested on questions with fixed wording. But the same question can be written in different ways while keeping its meaning and correct answer unchanged.

Prior Work & Research Gap. Srikanth et al. (2024) found that rephrasing questions can affect model correctness. Testing only one version does not show how different linguistic changes affect answers across different reasoning tasks.

Our Study. We test how lexical, syntactic, and information-order changes affect model correctness across three reasoning tasks and three open-weight language models.

Research Questions & Hypothesis

RQ1 — Performance Change. How does model accuracy change with lexical, syntactic, and information-order transformations?

RQ2 — Correctness Stability. How often do correct answers become incorrect, or incorrect answers become correct, after rephrasing?

RQ3 — Robust Accuracy. How often do models answer correctly across all four versions compared with the original?

Hypothesis. Rephrasing a question can change model correctness even when its meaning and correct answer stay the same. We expect the effect to vary across transformation types.

Contribution & Study Scope

We test three types of linguistic changes across three reasoning tasks using three open-weight language models.

- **Datasets:** CommonsenseQA, GSM8K, LogiQA
- **Transformations:** Lexical, Syntactic, Information Order
- **Models:** Qwen2.5-3B, Llama-3.2-3B, Gemma-3-1B

30 questions × 4 versions × 3 models = 360 responses

Main Contribution. Our **paired, multi-domain evaluation** compares original questions with three meaning-preserving versions. We measure **accuracy**, **correctness flips**, and **robust accuracy** to assess how rephrasing affects model correctness.

Experimental Example

CommonsenseQA — Picture Frame

Original: Where can you put a picture frame when it's not hung vertically?

Lexical: Where can you place a picture frame when it is not hanging vertically?

Syntactic: When a picture frame is not hung vertically, where can it be placed?

Info. Order: A picture frame is not being hung vertically. Where can you put it?

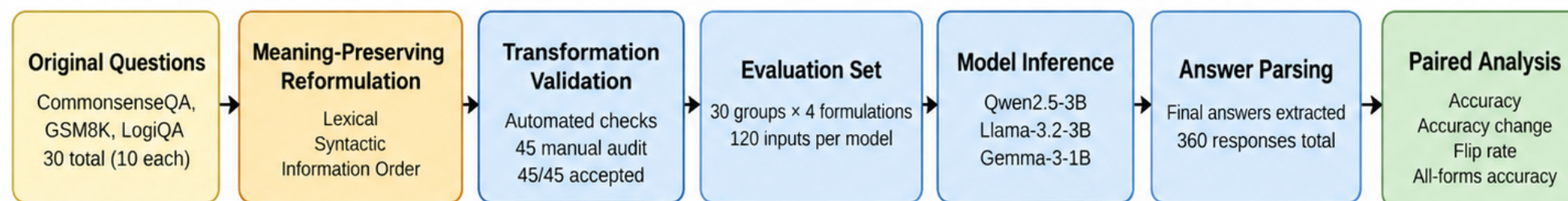
Correct Answer: E. Table

Observation: All four versions preserve the intended answer while changing how the question is expressed.

Methodology & Experimental Design

Dataset Selection. We selected **30 original questions**, with 10 from each dataset: **CommonsenseQA**, **GSM8K**, and **LogiQA**. These cover commonsense, mathematical, and logical reasoning, allowing us to examine how models respond across different reasoning tasks.

Experimental Setup. We tested each question in four versions: **Original**, **Lexical**, **Syntactic**, and **Information Order**. Each reformulation was designed to keep the original meaning, reasoning requirements, and correct answer.



Transformation Types. Lexical changes replace words or phrases, Syntactic changes modify sentence structure, and Information Order rearranges existing facts. All three aim to preserve the question's original meaning.

Paired Evaluation. Each original question and its three rewritten versions form one question group. We tested all four versions on each model using the same inference settings. This allows us to compare accuracy and correctness changes within each group, rather than across unrelated questions, and examine the effect of each transformation.

Transformation Quality Control

All **90 transformations** passed structural checks to confirm that answer choices and numbers were preserved (where applicable) and the text had changed.

We manually reviewed a stratified sample to check whether the original meaning and correct answer were preserved and the transformations were valid.

45 / 45 accepted

None of the reviewed transformations changed the correct answer, removed important information, or introduced new facts. These checks reduce the risk of attributing transformation errors to model behavior.

Models & Evaluation

Models. Three open-weight instruction-tuned LLMs: Qwen2.5-3B-Instruct, Llama-3.2-3B-Instruct, and Gemma-3-1B-IT from distinct model families.

Inference. Each model receives **120 inputs** with deterministic decoding (do_sample=False, max_new_tokens=32). Only final answers are scored.

Evaluation Metrics.

- **Accuracy:** Percentage of correct answers.
- **Accuracy Change:** Transformed minus original accuracy.
- **Flip Rate:** Percentage of correctness changes (Correct↔Wrong).
- **Robust Accuracy:** Question groups correct in all four formulations. Consistency is measured through correctness, not identical answer text.

Statistics & Reproducibility

We used exact **McNemar tests**, question-group **bootstrap 95% CIs**, and **Benjamini–Hochberg correction**.

Code, data, validation, model outputs, and analysis:
github.com/Yash-Gavade/ReasonShift



Main Result

Qwen: 63.3% → 30.0%

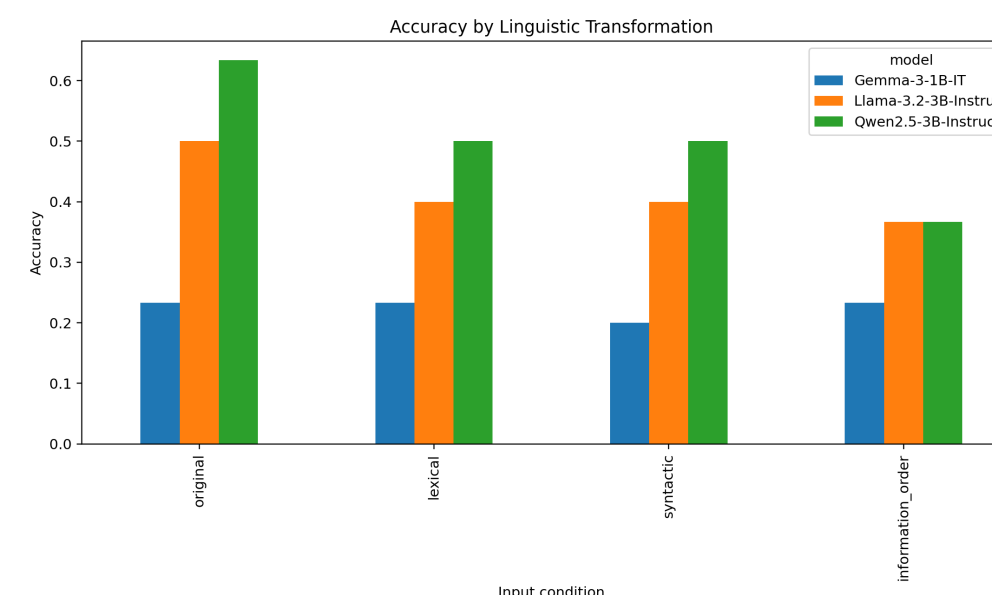
Original Accuracy → Robust Accuracy

Difference: −33.3 pp

Model	Original	Robust	Gap
Qwen	63.3%	30.0%	−33.3 pp
Llama	50.0%	26.7%	−23.3 pp
Gemma	23.3%	20.0%	−3.3 pp

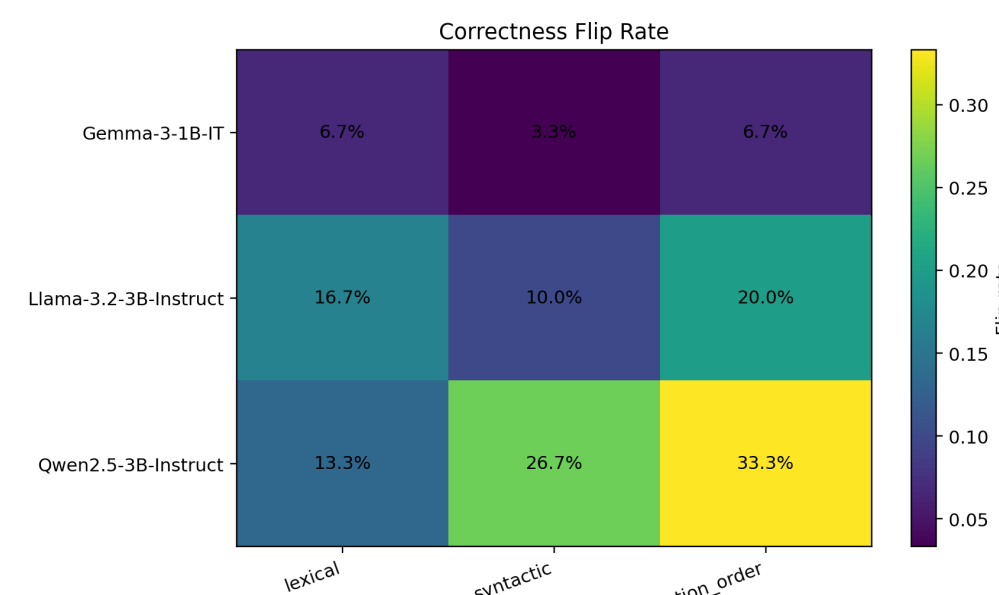
Qwen showed the largest gap (33.3 pp) between original and robust accuracy. Robust accuracy requires correct answers across all four versions.

Results by Transformation



Accuracy by Transformation

Qwen: Original → Info. Order
63.3% → 36.7%



Correctness Flip Rate

Qwen: Information Order
33.3% (9 C→W, 1 W→C)

For Qwen, changing the information order resulted in 8 fewer correct answers out of 30.

Findings & Hypothesis

- **Answers changed after rephrasing.** Some correct answers became incorrect, while some incorrect answers became correct.
- **Original accuracy did not tell the full story.** All three models had lower robust accuracy, showing that testing only one version of a question can miss changes in correctness.
- **Results differed by transformation type.** For Qwen, Information Order showed the largest accuracy drop and highest correctness flip rate.

These results support our hypothesis descriptively. However, none of the comparisons remained statistically significant after correction.

Discussion & Conclusion

Statistical Interpretation. For Qwen with Information Order, the exact McNemar test gave $p = 0.0215$. After Benjamini–Hochberg correction, $p_{BH} = 0.7734$.

No comparison remained statistically significant after correction.

Limitations. Our study used **30 question groups**, three small open-weight models, and three transformation types, limiting statistical power and generalizability. Only **45/90 transformations** were manually checked for meaning preservation.

Conclusion. Rephrasing a question can change whether a model answers correctly. Testing multiple versions reveals differences that a single version may miss. Larger studies with fully validated transformations are needed.